

BMIA - Practical case 1: Methodology for Diabetic Macular Edema classification

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I. INTRODUCTION

The goal of this practice is to implement a methodology for building an automatic classifier that given an Optical Coherence Tomography (OCT) image was able to detect whether there is the presence of Diabetic Macular Edema (DME) or not. The image set provided consists of 100 images in "jpg" format, 50 of them with presence of DME and 50 without it. Images have different dimensions, all of them with rectangular shapes.

II. METHODOLOGY

The process and chosen techniques for the current methodology were divided into 4 parts: image pre-processing, texture-feature extraction, feature selection and the image classifier. Each section is explored below:

A. Pre-processing of the image

In order to homogenize all the images, a small pre-processing was done. The transformations followed were:

- Converting to grey colour space.
- Resizing to 512 x 1024 with a linear interpolation from cv2 package.
- Cropping for removing a small note in the bottom left corner (Figure 1): y-axis between 0 and 450, and x-axis between 100 and 1024.

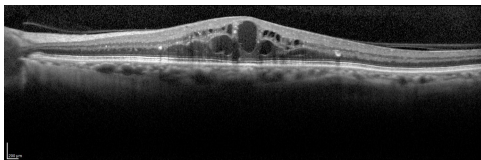


Fig. 1. Original image

B. Features extraction

Once all images had the same size and the region of interest was gotten, the next step was to extract features of the images for later to be the input of the classifier.

Here, various feature extraction methods based on texture such as Local Binary Pattern (LBP), Principal Component Analysis (PCA), Gray Level Co-occurrence Matrix (GLCM)

and Gabor filters; and intensity-based features as Histogram of Oriented Gradients (HOG) were used.

1) *Local Binary Pattern (LBP)*: Local Binary Patterns is a very efficient texture operator which labels the pixels of an image by thresholding the neighbourhood of each pixel and considers the result as a binary number. LBP descriptors efficiently capture the local spatial patterns and the grey scale contrast in an image.

Input parameters for obtaining the LBP were:

- *numPoints* = 28
- *radius* = 6
- *method* = "uniform"

After computing the histogram, over the cell, of the frequency of each "number" occurring, 20 features (histogram bins) were resultant.

2) *Grey Level Co-occurrence Matrix (GLCM)*: A statistical method of examining texture that considers the spatial relationship of pixels is the Grey Level Co-occurrence Matrix, also known as the Grey Level spatial dependence matrix. The GLCM functions characterize the texture of an image by calculating how often pairs of pixel with specific values and in a specified spatial relationship occur in an image, creating a GLCM, and then extracting statistical measures from this matrix.

Input parameters for obtaining the matrix were:

- *distances* = 1
- *angles* = 0
- *grey level* = 256

The properties gotten for the matrix were: contrast, dissimilarity, homogeneity, energy, correlation and ASM (Angular Second Model).

3) *Principal Component Analysis*: PCA is the best-known unsupervised linear feature extraction algorithm; it is a linear mapping which uses the eigenvectors with the largest eigenvalues. Twenty features were resultant from the Principal Component Analysis.

4) *Gabor filters*: this filter is a linear filter used for edge detection. Frequency and orientation representations of Gabor filters are similar to those of the human visual system, and they have been found to be particularly appropriate for texture representation and discrimination. In the spatial domain, a 2D Gabor filter is a Gaussian kernel function modulated by a sinusoidal plane wave. Thus, image analysis with Gabor filters is thought to be similar to perception in the human visual system.

A battery of different kernels was used; input parameters for creating the combinations were:

- $\gamma = 0.5$
- $\sigma = 0.56$
- $\phi = 0$
- $\theta = [0, \pi, 2\pi, 3\pi, 4\pi]$
- $\lambda = [2\pi, 4\pi, 6\pi, 8\pi, 10\pi]$

Finally, for all the filters applied, fifty features were gotten: 25 with the mean amplitude and 25 with local energy for each convolved image.

5) *Histogram of Oriented Gradients (HOG)*: The technique counts occurrences of gradient orientation in the localized portion of an image. The HOG descriptor focuses on the structure or the shape of an object. It is better than any edge descriptor as it uses magnitude as well as angle of the gradient to compute the features. For the regions of the image it generates histograms using the magnitude and orientations of the gradient.

After getting the results from the HOG process, a PCA with 20 was done in order to reduce and extract the relevant information from the HOG matrix.

At the end of this process, the set of features getting from the images was composed for 126 features (Table I).

TABLE I
NUMBER OF FEATURES EXTRACTED

Technique	Features
Principal Component Analysis	20
Local Binary Pattern	26
Gabor filter	50
Grey Level Co-occurrence Matrix	6
Histogram Oriented Gradient	20

C. Feature selection

The purpose of feature selection is to determine a subset within the set of features in order to minimize the classification error based on various criteria. A straightforward method of feature selection is Recursive Feature Elimination (RFE) where, given an external estimator that assigns weights to features, the goal of RFE is to select features by recursively considering smaller and smaller sets of features.

As an estimator, a RandomForest with 200 estimators was used and 30 features were selected. In Table II the selected features after the Recursive Feature Elimination are shown.

TABLE II
NUMBER OF FEATURES SELECTED

Technique	Features	Features selected
Principal Component Analysis	20	11
Local Binary Pattern	30	4
Gabor filter	50	10
Grey Level Co-occurrence Matrix	6	2
Histogram Oriented Gradient (HOG)	20	3

D. DME classifier

Several experiments with different combinations of parameters and algorithms were applied, finally, the model selected was the Random Forest. This method combines ensemble learning methods with the decision tree framework to create multiple randomly drawn decision trees from the data, averaging the results to output a new result that often leads to strong predictions.

As the first step, the dataset was split into a train (70 %) and a test set (30%). Also, a validation set was used when using a k-fold function, the modelling processes was:

- 1) K-Fold cross-validation with 10 folds, using stratified method.
- 2) Checking the stability of the metric f1 score.
- 3) Training the model using the whole train set.
- 4) Prediction with the test set.

The final results of the model chosen were:

- F1 score mean 10 k-fold: 0.787
- F1 score std 10 k-fold: 0.135

In Figure 2, the confusion matrix over the test set is shown; only one of the images was classify wrongly and the f1 score in the final prediction was 0.971. It is important to point out that other experiments had slightly better results considering the f1 score mean and standard deviation, however, in test, the chosen one had the best result in test.

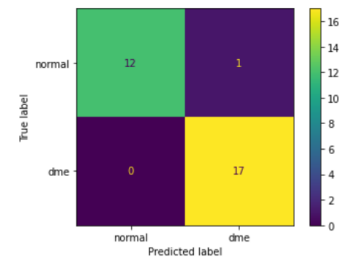


Fig. 2. Confusion Matrix

III. DISCARDED EXPERIMENTS

In Figure 3, the parameters and results of experiments done and discarded are shown:

Experiment	Train results	Test results
Hyperparameters chosen: - Algorithm: <i>RandomForestClassifier</i> (<i>n estimators</i> = 300) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 20) - RFE <i>n features</i>: 20 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.813 - F1 score std 10 k-fold: 0.150	0.938
Hyperparameters chosen: - Algorithm: <i>RandomForestClassifier</i> (<i>n estimators</i> = 300) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 20) - RFE <i>n features</i>: 30 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.826 - F1 score std 10 k-fold: 0.143	0.971
Hyperparameters chosen: - Algorithm: <i>GradientBoostingClassifier</i> (<i>n estimators</i> = 200, <i>learning rate</i> = 0.8, <i>max depth</i> = 2) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 20) - RFE <i>n features</i>: 20 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.842 - F1 score std 10 k-fold: 0.137	0.938
Hyperparameters chosen: - Algorithm: <i>GradientBoostingClassifier</i> (<i>n estimators</i> = 200, <i>learning rate</i> = 0.8, <i>max depth</i> = 2) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 20) - RFE <i>n features</i>: 30 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.901 - F1 score std 10 k-fold: 0.112	0.938
Hyperparameters chosen: - Algorithm: <i>GradientBoostingClassifier</i> (<i>n estimators</i> = 200, <i>learning rate</i> = 0.8, <i>max depth</i> = 2) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 20) - RFE <i>n features</i>: 40 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.788 - F1 score std 10 k-fold: 0.172	0.970
Hyperparameters chosen: - Algorithm: <i>GradientBoostingClassifier</i> (<i>n estimators</i> = 200, <i>learning rate</i> = 0.8, <i>max depth</i> = 2) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 20) - RFE <i>n features</i>: 40 - PCA <i>n components</i>: 40	- F1 score mean 10 k-fold: 0.777 - F1 score std 10 k-fold: 0.196	0.970
Hyperparameters chosen: - Algorithm: <i>RandomForestClassifier</i> (<i>n estimators</i> = 300), - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i> = 200) - RFE <i>n features</i>: 30 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.827 - F1 score std 10 k-fold: 0.160	0.944
Hyperparameters chosen: - Algorithm: <i>SVC</i> (<i>kernel</i>: 'rbf', <i>C</i>: 1.0) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i>: 20) - RFE <i>n features</i>: 20 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.721 - F1 score std 10 k-fold: 0.147	0.848
Hyperparameters chosen: - Algorithm: <i>SVC</i> (<i>kernel</i>: 'rbf', <i>C</i>: 10) - RFE estimator: <i>RandomForestClassifier</i> (<i>n estimators</i>: 20) - RFE <i>n features</i>: 20 - PCA <i>n components</i>: 20	- F1 score mean 10 k-fold: 0.725 - F1 score std 10 k-fold: 0.159	0.846

Fig. 3. Discarded experiments